University of BRISTOL

COMS20018, Introduction to AI: Sequential Decision Making

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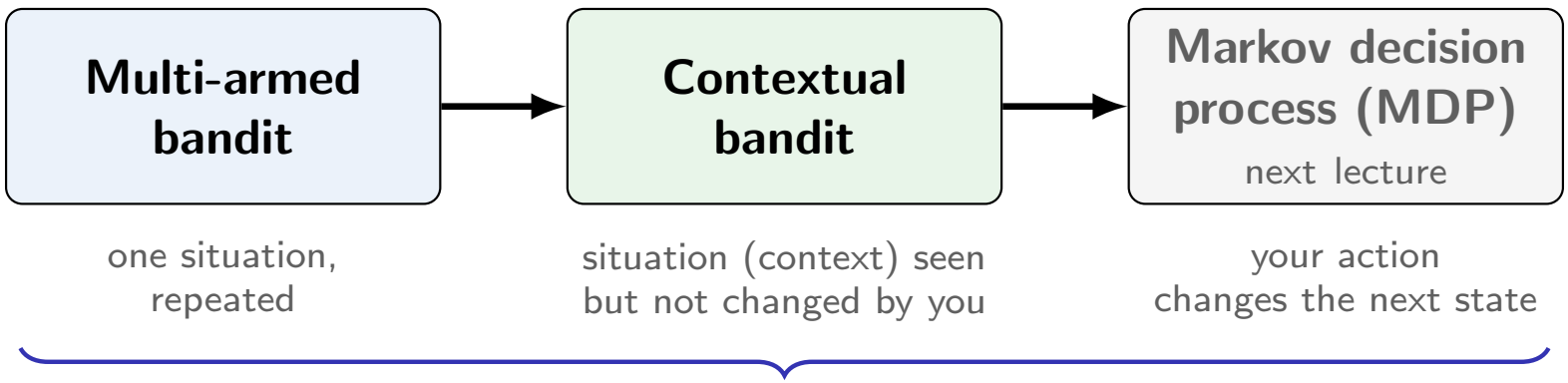
Agenda

- Motivation
- Multi-armed bandits
- Bandit algorithms
- Contextual bandits
- Off-policy evaluation via causal inference

From causes to decisions

L13 asked: *what would happen if I intervened?*

Today: *sequentially select actions (intervene) and learn from the results*.



$\Rightarrow$ We are never told the right answer, only the **reward** for what we actually did.

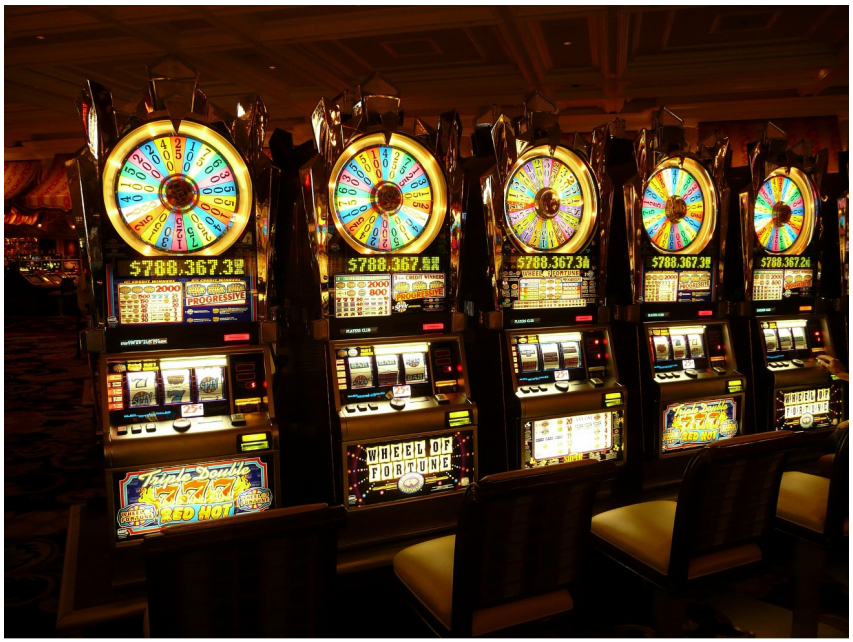
Part 1

Multi-armed Bandits

Explore or exploit?

Multi-armed bandits | Bandit algorithms | Contextual bandits | Off-policy evaluation

Let's play a game!



- Slot machines with **unknown** reward distributions.
- A limited number of coins.

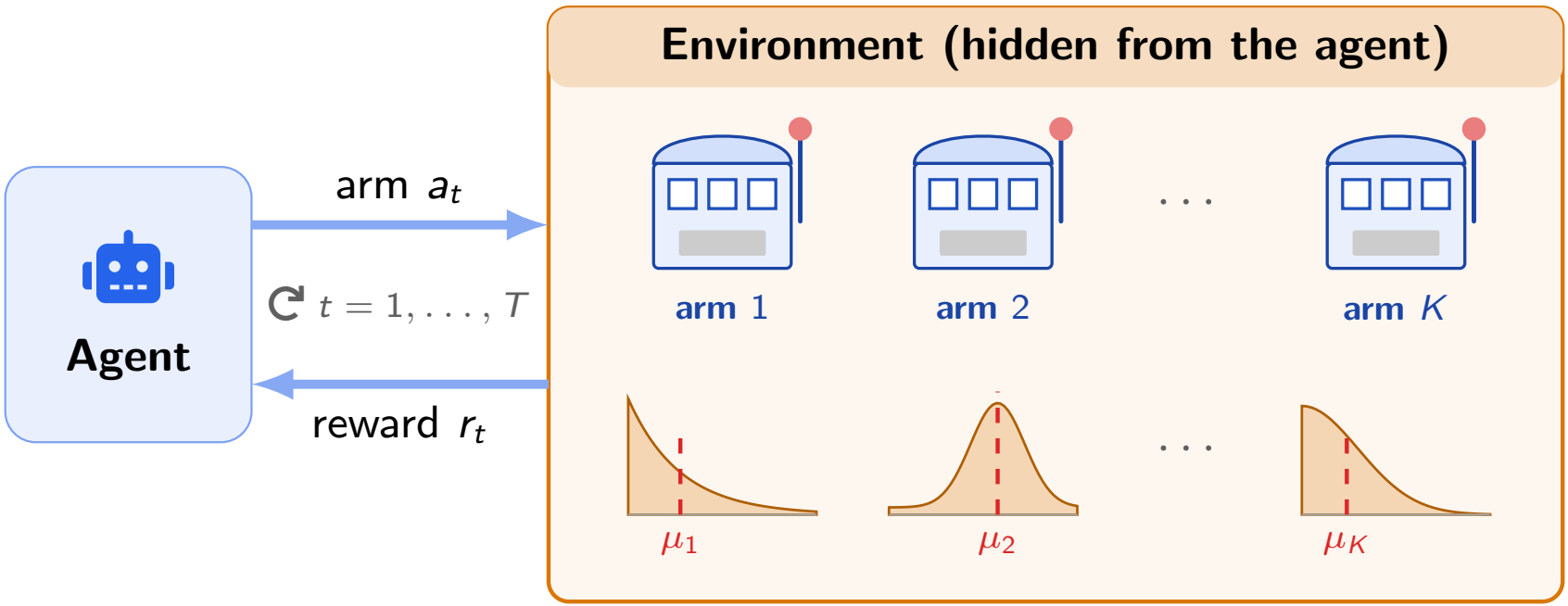
Goal: win as much money as possible.

Question

What's your strategy?

The multi-armed bandit problem

- **Multi-armed bandit**: K machines (**arms**), each with its own unknown reward distribution.



- K arms $a \in \{1, \ldots, K\}$, unknown means $\mu_1, \ldots, \mu_K$.
- Round $t = 1, \ldots, T$: pick arm a_t, observe reward $r_t \sim$ arm a_t's distribution.
- Goal: maximise **cumulative reward** $\sum_{t=1}^{T} r_t$.

Formal setup: maximise reward, minimise regret

Initialise $\hat{Q}(a)$ for every arm a
for $t = 1, \ldots, T$ **do**
  choose an arm a_t with policy π
  play a_t, observe reward r_t
  update $\hat{Q}(a_t)$ using r_t
end

$\hat{Q}(a) =$ **estimated value** of arm a: the average reward seen so far from arm a, our guess of μ_a.

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Expected cumulative regret: the cost of not knowing the best arm

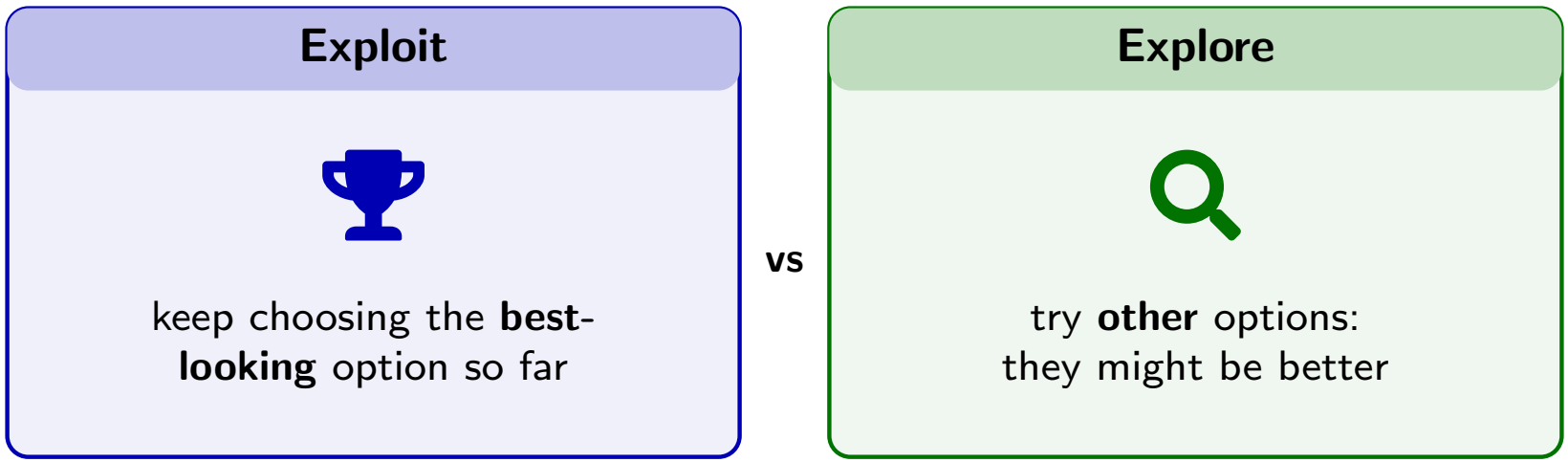
$$\mathbb{E}[R_T] = \mathbb{E}\Big[\sum_{t=1}^{T}(\mu^* - \mu_{a_t})\Big] = T\mu^* - \mathbb{E}\Big[\sum_{t=1}^{T} r_t\Big]$$

$\mu^* = \max_a \mu_a$; expectation because a_t depends on random rewards (and random exploration)

$\Rightarrow$ $T\mu^*$ is fixed: **max reward** $\Leftrightarrow$ **min regret**.

$\Rightarrow$ Goal: **sublinear regret**, $\mathbb{E}[R_T]/T \to 0$ as $T \to \infty$: we end up (almost) always playing the best arm.
Linear regret: we never stop making mistakes.

The exploration–exploitation dilemma



- Too much exploitation: may get stuck on a suboptimal arm.
- Too much exploration: rounds wasted on bad arms.

$\Rightarrow$ The central challenge of bandits, and of reinforcement learning more broadly.

See Sutton & Barto (2018), Chapter 2.

Part 2

Bandit Algorithms

How to design a policy?

Multi-armed bandits | **Bandit algorithms** | Contextual bandits | Off-policy evaluation

Running example: choosing a coffee shop



- 3 coffee shops near campus: A, B, C (the arms).
- Each visit: coffee is **great** ($r = 1$) or **okay** ($r = 0$).

	Shop A	**Shop** B	**Shop** C
μ_a (true, *hidden* from you)	0.3	0.5	0.7

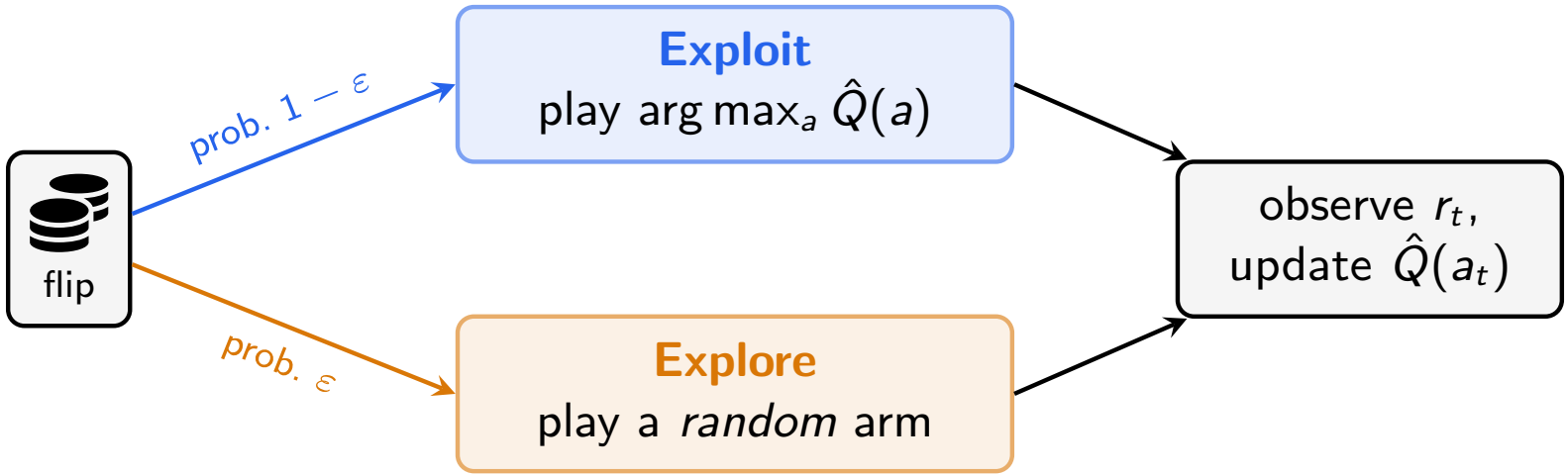
You don't know the μ_a: you only learn by visiting.

$\Rightarrow$ Visit shop B every day for $T = 100$ days:
$R_{100} = 100 \times (0.7 - 0.5) = 20$, about 20 great coffees lost.

$\Rightarrow$ Greedy can get stuck forever: we need to keep **exploring**.

ε-greedy

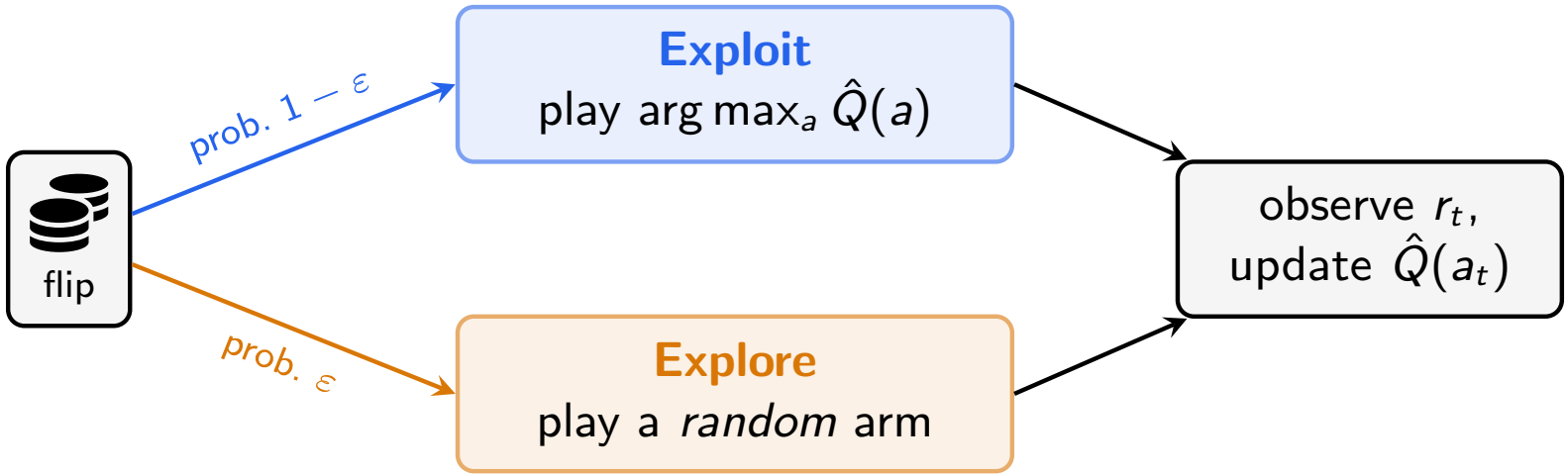
Fix a small $\varepsilon \in (0, 1)$, e.g. 0.1. Each round, flip a biased coin:



See Sutton & Barto (2018), Sections 2.2–2.4.

ε-greedy

Fix a small $\varepsilon \in (0, 1)$, e.g. 0.1. Each round, flip a biased coin:



$$N(a_t) \leftarrow N(a_t) + 1, \qquad \hat{Q}(a_t) \leftarrow \hat{Q}(a_t) + \frac{1}{N(a_t)}\left(r_t - \hat{Q}(a_t)\right)$$

running average of the rewards seen from arm a_t

- ε can **depend on time**: ε_t starts large (we know little, so explore a lot) and shrinks as the estimates improve, e.g. $\varepsilon_t = \min(1, c/t)$.

$\Rightarrow$ Every arm keeps being tried, so a lucky or unlucky start cannot lock us in, unlike greedy.

See Sutton & Barto (2018), Sections 2.2–2.4.

Upper Confidence Bound (UCB)

- ε-greedy explores *blindly* (uniformly at random).
- **UCB**: "optimism in the face of uncertainty".
- Track $\hat{Q}(a)$ (estimated value) and $N(a)$ (times played).

$$a_t = \arg\max_a \underbrace{\hat{Q}(a)}_{\text{estimated value}} + \underbrace{c\sqrt{\frac{\ln t}{N(a)}}}_{\text{uncertainty bonus}}$$

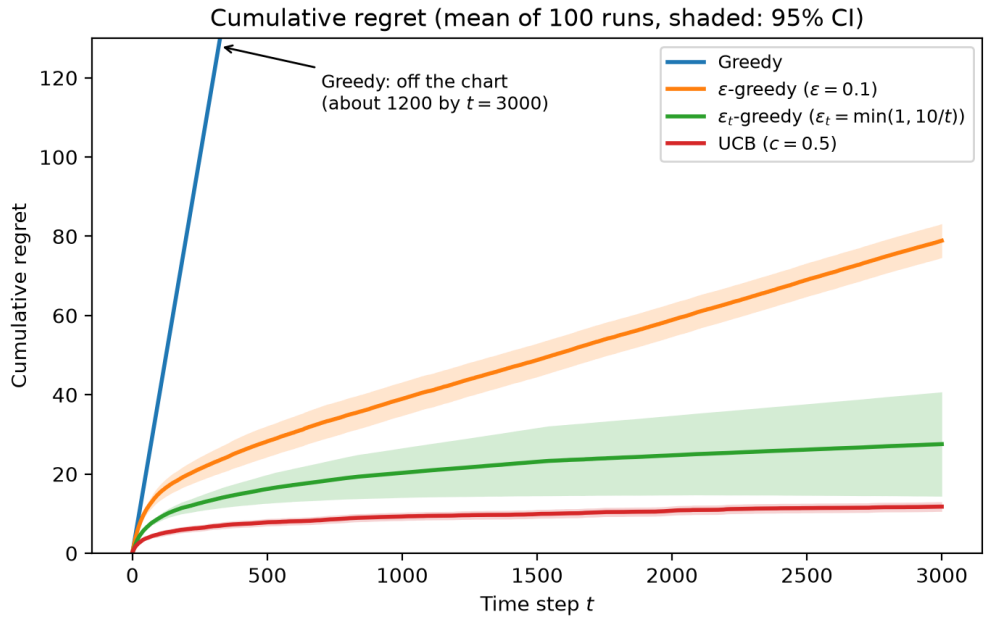
- Bonus large when $N(a)$ small; shrinks as $N(a)$ grows. $c > 0$ controls exploration.

$\Rightarrow$ Explores the arms we are most **uncertain** about, not at random.

See Sutton & Barto (2018), Section 2.7.

Does it work? Comparing the policies

Setup: the coffee shops, $\mu = (0.3, 0.5, 0.7)$, reward 1 (great) or 0 (okay); $T = 3000$ days; mean of 100 runs, shaded: 95% CI.



- **Greedy**: *linear* regret: stuck on a suboptimal shop.
- ε-**greedy**: still *linear*: explores 10% of the time, forever.
- ε_t-**greedy**, **UCB**: regret *flattens* (sublinear).

$\Rightarrow$ UCB is best and most reliable; ε_t sometimes locks onto a worse shop.

See Sutton & Barto (2018), Chapter 2.

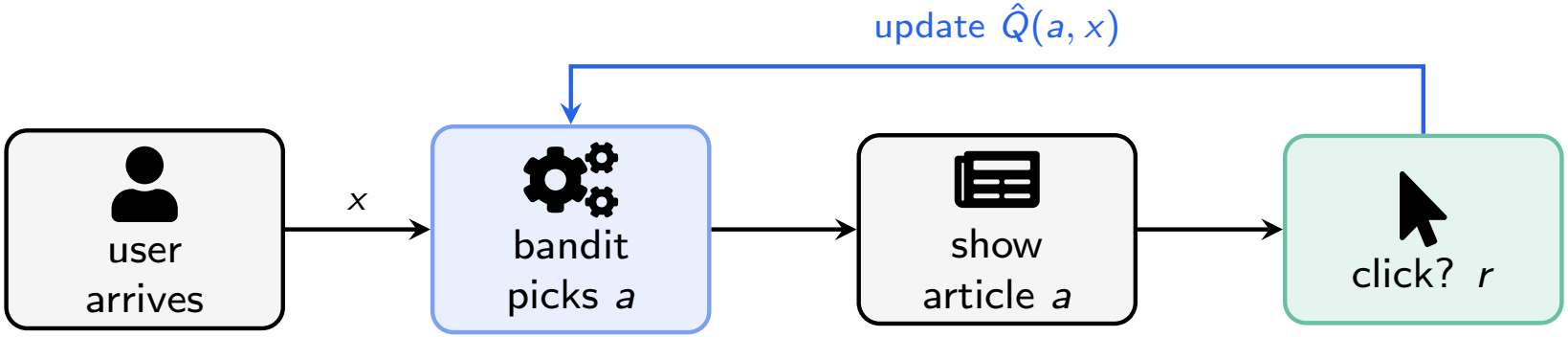
Part 3

Contextual Bandits

The best action depends on who you ask

Multi-armed bandits | Bandit algorithms | **Contextual bandits** | Off-policy evaluation

Example: personalised news recommendation



- **Context** x: the user's interests (sport fan or tech fan).
- **Arms**: Sport, Tech or Politics article.
- **Reward**: click ($r = 1$) or not.

$\Rightarrow$ No single article is best for everyone: using x lets the bandit learn what works *for each kind of user*.

Contextual bandits

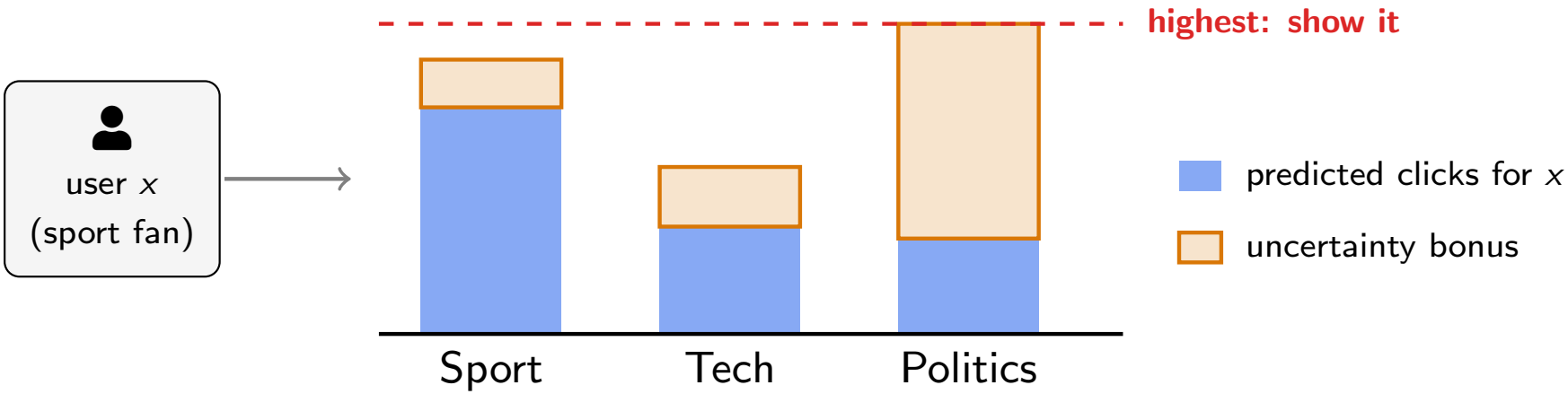
- Each round t:
  1. Environment reveals a **context** x_t (e.g. user features).
  2. Agent chooses arm a_t, *depending on* x_t.
  3. Environment reveals reward r_t (for that arm only).
- Expected reward now depends on the context: $\mathbb{E}[r \mid a, x]$.
- A **policy** maps contexts to arms: $a_t = \pi(x_t)$.

$\Rightarrow$ Same explore/exploit problem, but $\hat{Q}(a)$ becomes $\hat{Q}(a, x)$: "how good is arm *a for this context*?"

See Sutton & Barto (2018), Section 2.9 ("associative search").

From UCB to LinUCB

- **LinUCB** = UCB + a **linear model** of the context: show the article with the highest *prediction* + *bonus*.



Illustrative numbers.

- **Prediction**: $\hat{Q}(a, x) = x^\top \hat{\theta}_a$, a linear regression on user features x, one per article (L02–L03).
- **Bonus**: large if a was rarely shown to users *like* x (here: Politics).

See Li, Chu, Langford, Schapire (2010), WWW.

Part 4

Off-Policy Evaluation

Is a new policy better, before we deploy it? (causal inference, L13)

Multi-armed bandits | Bandit algorithms | Contextual bandits | **Off-policy evaluation**

Case study: Yahoo! front page news

- **Setting**: Yahoo! Front Page "Today Module": which article from an editor-curated pool to highlight at the STORY position, for each visitor.
- **Context**: user features (demographics, browsing history) and article features.
- **Data**: >33 million visits (May 2009).



Today Module (Li et al., 2010, Fig. 1).

⇒ Using context gave +12.5% clicks over a context-free bandit; the gap was largest when data were scarce.

See Li, Chu, Langford, Schapire, "A contextual-bandit approach to personalized news article recommendation", WWW 2010 (algorithm: LinUCB).

The off-policy evaluation problem at Yahoo!

- Goal: how well would a new policy π (e.g. LinUCB) do?
- Running π live is **infeasible**: we only have **logs** from an **old policy** π_0.
- Clicks are seen only for π_0's articles, which often differ from π's.

User	Old policy shown	Click	π would show	Click under π
sport fan	Sport	1	Sport	1
tech fan	Politics	0	Tech	?
sport fan	Tech	0	Sport	?
tech fan	Tech	1	Tech	1

Illustrative log.

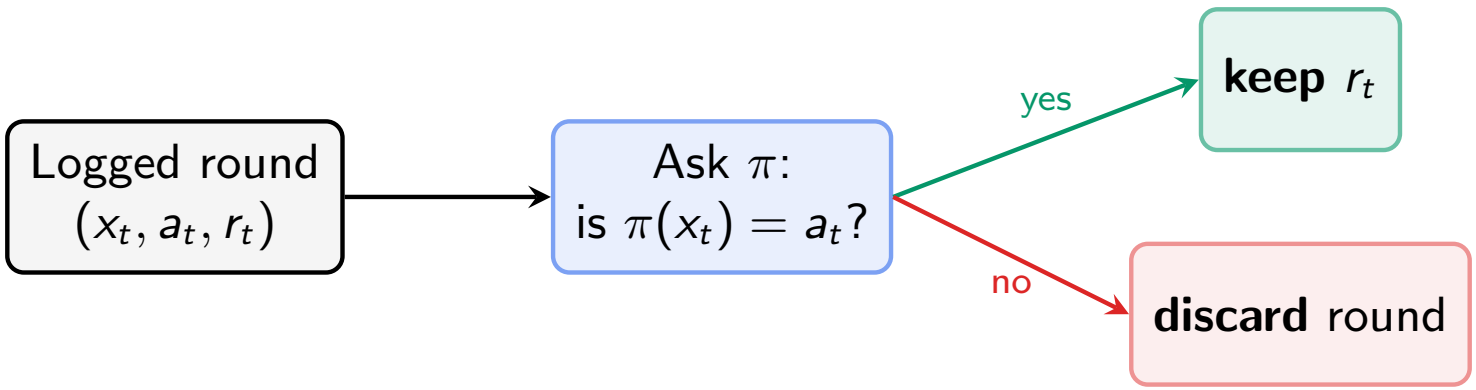
$\Rightarrow$ Each ? is a **counterfactual** (L13): this is **off-policy evaluation**.

$\Rightarrow$ π_0 chose based on the user: **confounded**. Fix: log some traffic **at random**.

See Li, Chu, Langford, Schapire (2010), WWW, Section 4.

The replay method

- Take the **random** log and replay it, one round at a time:



$\hat{V}(\pi)$ = average reward over the **kept** rounds

$\Rightarrow$ Works because the random log is a **randomised experiment**.

$\Rightarrow$ Cost: only about $1/K$ of the rounds are kept.

See Li, Chu, Langford, Wang (2011), WSDM.

Replay: a worked example

New policy π: sport fan $\rightarrow$ Sport, tech fan $\rightarrow$ Tech.

Round	User	Shown	Click	π shows	Keep?
1	sport fan	Sport	1	Sport	✓
2	sport fan	Politics	0	Sport	✗
3	tech fan	Tech	1	Tech	✓
4	tech fan	Sport	0	Tech	✗
5	sport fan	Sport	0	Sport	✓
6	tech fan	Tech	1	Tech	✓

Illustrative numbers.

$\Rightarrow$ Keep 1, 3, 5, 6: $\hat{V}(\pi) = 3/4 = 0.75$ (random logger: 0.5).

When the log is not random, replay is biased

New policy π: show **Sport** to everyone. 100 sport fans, 100 tech fans.
Assume: if shown Sport, **50%** of sport fans click, **10%** of tech fans click.

True value: if we deployed π

	Users	Shown Sport	Clicks
sport fans	100	100 (100%)	50
tech fans	100	100 (100%)	10

π shows Sport to *all* users.

$$V(\pi) = \frac{\text{clicks}}{\text{users}} = \frac{50+10}{100+100} = 0.30$$

Replay: from the old policy π_0's log

	Users	Shown Sport	Clicks
sport fans	100	90 (90%)	45
tech fans	100	10 (10%)	1

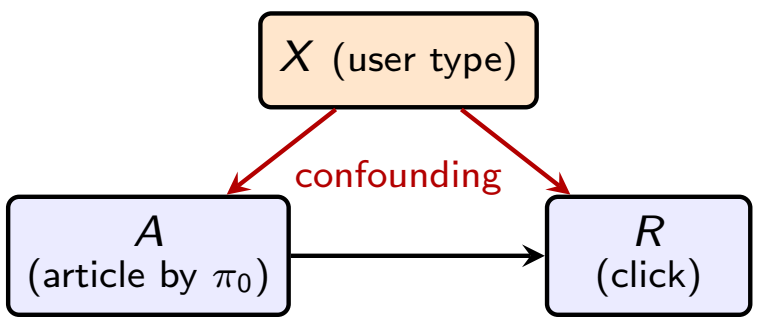
Replay keeps only the rounds shown Sport.

$$\hat{V}(\pi) = \frac{\text{clicks}}{\text{kept}} = \frac{45+1}{90+10} = 0.46 \quad \times$$

Illustrative numbers.

$\Rightarrow$ Kept rounds are 90% sport fans, not 50%: π_0's choice depends on the user, so the log is **confounded**.

How π_0 generated the log (x_t, a_t, r_t)



Inverse propensity scoring (IPS)

- **Propensity** $\pi_0(a \mid x)$: the probability that the old policy showed article a to user x.
- Idea: weight each kept round by $1/\pi_0(a_t \mid x_t)$: a rare choice stands in for the many rounds the old policy did not take.

$$\hat{V}_{\mathrm{IPS}}(\pi) = \frac{1}{n} \sum_{t=1}^{n} \frac{\mathbb{I}[\pi(x_t) = a_t]}{\pi_0(a_t \mid x_t)} r_t$$

Example: $\dfrac{45 \times \frac{1}{0.9} + 1 \times \frac{1}{0.1}}{200} = \dfrac{50 + 10}{200} = 0.30$ ✔

$\Rightarrow$ Needs **positivity**, $\pi_0(a \mid x) > 0$; tiny π_0 gives huge weights and high variance.

$\Rightarrow$ Random log ($\pi_0 = 1/K$): all weights equal, as in replay.

See Neal (2020), Section 7.6 (inverse probability weighting).

Bandits are everywhere



Clinical trials
which treatment
is most effective?



Online advertising
which ad gets clicked?



A/B testing
which website
design works best?



Dynamic pricing
which price
maximises revenue?



Recommender systems
which article,
product or song?



Routing & allocation
which route or server?

$\Rightarrow$ Repeated choices under uncertainty, with feedback only for the choice you made.

Summary

- **Multi-armed bandits**
  - Choose arms sequentially; only see the reward of the arm played.
  - Max cumulative reward $\Leftrightarrow$ min *regret*; aim for *sublinear* regret.
  - Core challenge: **exploration vs. exploitation**.
- **Bandit algorithms** (policies)
  - ε-greedy: explores at random; let ε_t decay.
  - UCB: explores where it is most *uncertain*.
- **Contextual bandits**
  - The best arm depends on the context x: learn $\hat{Q}(a, x)$.
  - LinUCB: UCB + a linear model of the context.
- **Off-policy evaluation**
  - A policy's value is causal: $\mathbb{E}_X[\mathbb{E}[R \mid do(A{=}\pi(X)), X]]$.
  - From logs: **replay** (random logs) or **IPS** (non-random logs).

Reading

- Sutton & Barto, *Reinforcement Learning: An Introduction* (2nd ed.), Chapter 2 (bandits). Free online.
- Li, Chu, Langford, Schapire. "A Contextual-Bandit Approach to Personalized News Article Recommendation." WWW 2010 (Yahoo! case study).
- Li, Chu, Langford, Wang. "Unbiased Offline Evaluation of Contextual-Bandit-based News Article Recommendation Algorithms." WSDM 2011 (replay method).
- Neal, *Introduction to Causal Inference* (2020), Chapter 5 and Section 7.6 (inverse probability weighting). Free online.
- Further reading: Lattimore & Szepesvári, *Bandit Algorithms* (2020), Chapters 1–2. Free online.